

Novel Gaussian Beam Method for the Rapid Analysis of Large Reflector Antennas

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Abstract—A relatively fast and simple method utilizing Gaussian beams (GBs) is developed which requires only a few seconds on a workstation to compute the near/far fields of electrically large reflector antennas when they are illuminated by a feed with a known radiation pattern. This GB technique is fast, because it completely avoids any numerical integration on the large reflector surface which is required in the conventional physical optics (PO) analysis of such antennas and which could take several hours on a workstation. Specifically, the known feed radiation field is represented by a set of relatively few, rotationally symmetric GBs that are launched radially out from the feed plane and with almost identical interbeam angular spacing. These GBs strike the reflector surface from where they are reflected, and also diffracted by the reflector edge; the expressions for the fields reflected and diffracted by the reflector illuminated with a general astigmatic incident GB from an arbitrary direction (but not close to grazing on the reflector) have been developed in [1] and utilized in this work. Numerical results are presented to illustrate the versatility, accuracy, and efficiency of this GB method when it is used for analyzing general offset parabolic reflectors with a single feed or an array feed, as well as for analyzing nonparabolic reflectors such as those described by ellipsoidal and even general shaped surfaces.

Index Terms—Gaussian beams, reflector antennas.

I. INTRODUCTION

A FAST procedure based on a novel application of Gaussian beams (GBs) is described in this paper for predicting the near and far fields of electrically large reflectors, with slowly varying surface properties, when they are illuminated by a given feed antenna or a feed array. The conventional procedure for computing the fields of reflectors requires a numerical integration of the physical optics (PO) integral over the reflector surface; this can become computationally very slow and highly inefficient for large reflectors. Computational speed is particularly essential when performing reflector antenna synthesis wherein the reflector radiation pattern needs to be computed during each iterative step of the synthesis algorithm until the desired radiation pattern is obtained to within some prescribed bounds. Thus, in such a synthesis procedure, the numerical PO method may become nearly intractable for large reflectors. The present GB procedure for predicting the near and far fields of the reflector an-

tenna completely avoids any time-consuming numerical PO integration on the reflector surface. As a result, the present method is extremely fast as will be demonstrated for the analysis of large reflector antennas. A wide variety of applications of this GB method are presented here to illustrate its efficiency, versatility, and accuracy by analyzing not only offset parabolic reflectors with a single feed or an array feed, but also nonparabolic reflectors made up, for example, of ellipsoidal and even more general shaped surfaces.

Specifically, the electromagnetic (EM) radiation from a known feed, or from a known feed array, is represented by a set of relatively few GBs. The number of GBs required and the waists of the GBs at their launch points, as well as their axial direction at launch, are selected according to some simple criteria. An efficient point matching procedure, or a least squares method, can be employed for accurately determining the GB expansion coefficients (i.e., the initial amplitudes of the GBs at their points of launching). Once these GBs are launched from a single point, or in special cases from a very small number of points in the feed plane, they strike the reflector surface where they undergo reflection and diffraction by the surface of the parabolic reflector and its edge, respectively. Explicit, closed-form expressions for the fields associated with the reflection and diffraction of any given incident GB by a smooth reflector containing an edge, which have been derived in detail in a previous paper [1], are employed here. It is important to note that the closed-form expressions for the fields associated with the reflection and diffraction of any single GB which is incident on the reflector have been developed in [1], via an asymptotic evaluation of the corresponding radiation integral that is approximated therein by PO, for the single GB illumination case. Thus, the GB analysis presented here for predicting the total reflector near/far fields provides an asymptotic closed-form analytical solution of the PO integral for reflectors; this is in contrast to the conventional numerical evaluation of the PO integral. Indeed, the present closed-form PO-based GB analysis is shown here to have essentially the same level of accuracy as the numerical PO solution. It is well known that the PO approximation can be improved by using an edge diffraction correction based on the physical theory of diffraction (PTD) [2], [26], [3]; such a correction can be incorporated in a straightforward manner. Clearly, since the present GB analysis actually represents a closed-form solution of the PO integral itself, this GB analysis can thus be corrected by simply adding to it the exact same PTD correction as is done for the conventional solution based on a numerical evaluation of the PO integral. In particular, the PTD edge diffraction correction can be added to the PO solution as a

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line integral contribution around the reflector edge [4]–[10], [27]. Such a PTD correction provides an improvement to the PO based analysis primarily in regions of lower field levels. In the case of reflector antennas the PTD thus provides some improvement in the wide-angle sidelobe region. As an alternative to the PTD-based correction to PO, one can first employ the present PO-based GB solution in the region of the main beam and first few sidelobes, and then switch to the uniform geometrical theory of diffraction (UTD) [10]–[13] ray solution in the region of the intermediate and far-out sidelobes of reflector antennas. One notes that the UTD ray solution fails in regions of focused or nearly focused fields such as within the main beam regions of parabolic reflectors with a feed located at or near the reflector focus. Thus, the common approach for avoiding this difficulty of the UTD ray approach in focused regions of the radiated fields is to utilize the PO approximation in those regions where UTD fails; however, because the PO integral is evaluated numerically over the large reflector surface, it thus becomes highly inefficient as indicated earlier. The UTD approach, on the other hand, requires no integration on the reflector, and hence it is very efficient in its regions of validity. It is seen that the present GB approach retains all the advantages of the UTD ray method, and it has the important added advantage that it also remains valid in regions of focused or nearly focused fields where UTD fails. In fact, the present PO-based GB method may be viewed as a UTD for GBs which is analogous to UTD for rays except that the rays are now replaced here by GBs (where the GBs, unlike rays, remain valid in regions of focused fields). A PTD correction to this PO-based GB method, or alternatively switching from this GB to a UTD analysis outside the main beam region, can be accomplished in a straightforward fashion to improve the PO-based GB method as mentioned above; however, these have not been included in the present work.

The use of GBs as EM field basis functions has been studied and demonstrated by several authors in the past, notably in [14]–[18] to represent the fields of aperture antennas. The Gabor expansion was used therein to represent the known aperture field with Gaussian functions as basis elements in that expansion. The field of each of these Gaussian elements evolved from the aperture to the external region thereby directly providing the near and far fields of the aperture in closed form. The GB expansion employed here has some similarities to that in [14]–[16] but at the same time is also significantly different from it in the manner in which the GBs are chosen as will be discussed below. The aperture is discretized in [14]–[16] using a doubly periodic array of lattice points on which the Gaussian basis elements are centered. The fields external to the aperture consist of a set of GBs (where each GB is described within the paraxial region of the fields radiated by each Gaussian basis element). The GBs are launched radially outward from each of the periodic array of points that are chosen to discretize the aperture as in Fig. 1. Such a field representation constitutes a GB expansion in “phase-space.” When the period D_x and D_y of the lattice is specified in the $\hat{x}_f$ and $\hat{y}_f$ directions, then the Gabor expansion automatically provides information on the launching angle and the waist size of the rotationally symmetric GBs that emanate from

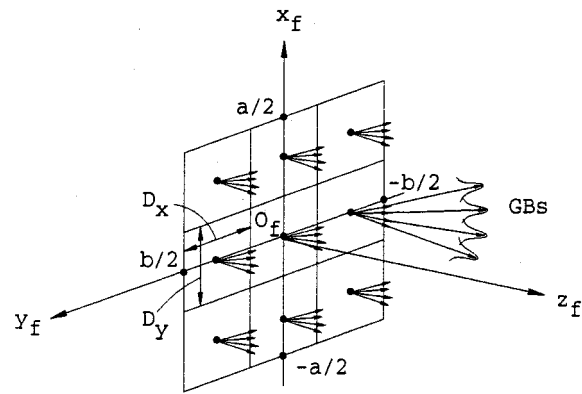


Fig. 1. GB expansion for the fields radiated from a periodic array of lattice points in the aperture defined over $|x_f| \leq a/2$; $|y_f| \leq b/2$; $z_f = 0$. The GBs are launched radially out from each lattice point in the periodic array. The lattice points may typically correspond to the phase center of each subaperture (of size $D_x \times D_y$).

each lattice point. While there are an infinite number of GBs launched from each of the finite number of N lattice points in [14]–[16], only the finite number of propagating GBs are significant outside the close neighborhood of the aperture; all the remaining GBs are evanescent and contribute negligibly except for points very close to the aperture. In practice, only a few evanescent GBs are typically needed in addition to the finite number of propagating GBs to accurately synthesize the original field distribution in the aperture. The coefficients of the GB expansion; i.e., the initial complex amplitudes of the GBs at launch, are found in [14]–[16] using the biorthogonality properties of the Gabor expansion. In [16], a Gabor-based GB expansion was also employed for analyzing the transmission of the fields of a planar aperture through plane and curved dielectric layers. However, previous work in using the Gabor representation was primarily restricted to two-dimensional (2-D) geometries in which the aperture was one-dimensional (1-D) (or linear) in extent; also, the diffraction of GBs was not included in those applications. An analysis of the reflector radiation pattern using a single GB to model the feed radiation was developed in [17] for the three-dimensional (3-D) problem, but the reflector edges were illuminated negligibly in that case also, and so the diffraction of the GB by the edges was excluded in [17]. An analysis of GB diffraction by a straight edge has been presented previously in [19]–[22]. It is noted that the work presented here is based on [1] which allows one to treat the substantially more general and practical problem of GB reflection and diffraction by a general curved surface which is truncated in a rather general but locally planar curved edge.

In the present application of the GB method, a procedure different from the Gabor expansion is utilized to analyze the fields radiated by a 3-D offset parabolic reflector that is illuminated by a realistic feed antenna as mentioned earlier. Furthermore, the feed illumination is such that the effects of diffraction by the reflector edge are important, as is true in most applications, and hence cannot be neglected. It is noted that a Gabor expansion could also have been utilized in the present GB approach to represent the fields incident on the reflector from the given

feed antenna. An advantage of the Gabor type expansion is that it guarantees completeness *a priori* and essentially removes any arbitrariness in the choice of initial beam parameters once the lattice spacing is chosen. However, the Gabor expansion gives rise to nonidentical GBs with nonuniform angular or interbeam spacing at launch thus providing less control on the choice of initial beam parameters; also, the coefficients of the GB expansion based on the Gabor representation generally require a relatively complicated numerical computation of biorthogonality integrals. Therefore, a somewhat different approach is employed here as it is very convenient for engineering applications. Specifically, the set of M GBs that are launched from each of the N lattice points of Fig. 1 in the present approach are such that they have the same interbeam spacing, $\Delta\theta$ (defined later) and, except for the different launching directions and polarization, all of the GBs are otherwise identical and rotationally symmetric. Relatively simple rules are developed for the choice of initial GB parameters in the present approach which also offer more control of these parameters. The computation of the GB coefficients (i.e., initial GB launching amplitudes) in the GB expansion are found through a relatively straightforward and efficient point-matching approach, or a least squares approach. It is important to note that in the present approach, the initial launching amplitudes of the GBs in the expansion for the feed radiation are chosen to match the far zone fields of the feed antenna which illuminates the reflector, rather than to match the feed antenna aperture field distribution. Consequently only the usual propagating GBs need to be included in the present expansion and no evanescent GBs are required.

It is noted that, for most realistic single feed reflector antenna applications, only one lattice point [$N = 1$ at O_f in Fig. 1, where O_f is the phase center of the feed in the feed coordinates (x_f, y_f, z_f)] is sufficient. For feed arrays, it may be necessary to have more than one lattice point if the reflector is not truly in the far zone of the entire array. For offset parabolic reflectors whose projected apertures range from about 40λ to 100λ in diameter (where λ = free space wavelength), it is seen in the present applications that the number M of GBs required is approximately 45 with $N = 1$ (corresponding to a single lattice point at the phase center, O_f of the feed antenna).

The format of this paper is as follows. In Section II, the general procedure for expanding the fields radiated by the feed antenna in terms of GBs is developed. The analysis of the near- and far-zone radiation pattern of fairly general offset reflector antennas is presented next in Section III specifically using the relatively simple closed-form expressions developed earlier in [1] for the reflection and diffraction of an arbitrary GB by a curved edge in a curved surface. Several numerical results based on this GB method are presented in Section IV for analyzing parabolic, ellipsoidal and even shaped reflectors; the computation of these results are shown to be extremely fast and accurate. In particular, it is noted that the present approach can provide the near and far fields of relatively large sized reflectors in a matter of only a few seconds on a workstation, whereas, the conventional numerical integration of the PO integral over the reflector surface requires several hours on the same workstation to furnish the same results. An $e^{+j\omega t}$ time dependence for the fields is assumed and suppressed.

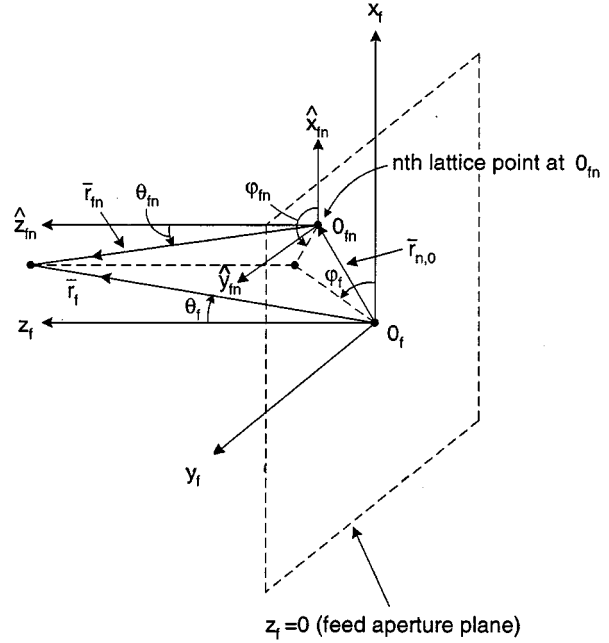


Fig. 2. Coordinate system (x_f, y_f, z_f) whose origin O_f is the phase reference for the feed. Also, coordinates $(x_{fnm}, y_{fnm}, z_{fnm})$ fixed in the m th GB launched from the n th lattice point in the $z_f = 0$ feed plane.

II. GB EXPANSION FOR THE FEED ANTENNA RADIATION

Let the feed antenna, which illuminates the reflector, radiate the electromagnetic fields $(\vec{E}_f^i, \vec{H}_f^i)$. One can express the fields in the far zone of the feed antenna as

$$\vec{H}_f^i(\vec{r}_f) \sim \frac{e^{-jk r_f}}{r_f} [\hat{\theta}_f F_\theta(\hat{r}_f) + \hat{\phi}_f F_\phi(\hat{r}_f)] \quad (1a)$$

$$\vec{E}_f^i(\vec{r}_f) \sim -Z_0 \hat{r}_f \times \vec{H}_f^i(\vec{r}_f) \quad (1b)$$

where the radiation pattern functions F_θ and F_ϕ in any $\hat{r}_f$ direction (with r_f, θ_f, ϕ_f as in Fig. 2) are known. Here k is the free space wavenumber and Z_0 is the free space impedance ($=Y_0^{-1}$).

One may represent $\vec{H}_f^i(\vec{r}_f)$ in (1a) in terms of a GB expansion, i.e., in terms of a sequence of GBs, by

$$\vec{H}_f^i(\vec{r}_f) \cong \sum_{n=1}^N \sum_{m=1}^M C_{nm} \vec{H}_{nm}(\vec{r}_{fnm}) e^{j\psi_{nm}} \quad (2)$$

where M propagating rotationally symmetric GBs are launched from each of the N lattice points in the feed plane which is the $z_f = 0$ plane as depicted previously in Fig. 1 such that the angular interbeam spacing $\Delta\theta$ [see (6)] between any pair of adjacent GBs is a constant. A typical n th lattice point at O_{f_n} where $n = 1, 2, 3, \dots, N$ is shown in Fig. 2. The direction $\hat{z}_f$ is perpendicular to the feed plane and it constitutes the feed boresight direction. Comments will be added later to deal with general rules for selecting the values of N and M . The magnetic field associated with each rotationally symmetric GB basis function

in (2) is given by

$$\begin{aligned} \bar{H}_{nm}(\bar{r}_{fnm}) &\approx (\bar{e}_{nm}) \frac{j b}{z_{fnm} + j b} \\ &\cdot \exp \left(-j k \left[z_{fnm} + \frac{1}{2} \frac{z_{fnm}}{z_{fnm}^2 + b^2} (x_{fnm}^2 + y_{fnm}^2) \right] \right) \\ &\cdot \exp \left(-\frac{1}{2} k b \frac{x_{fnm}^2 + y_{fnm}^2}{z_{fnm}^2 + b^2} \right) \end{aligned} \quad (3)$$

where $(x_{fnm}, y_{fnm}, z_{fnm})$ are local coordinates which are fixed in the m th propagating GB launched from the n th lattice point with $\hat{z}_{fnm}$ being the axial direction of that GB. The additional phase term $e^{j\psi_{fnm}}$ in (2) simply references the phase of $\bar{H}_{nm}$ launched from O_{f_n} to the feed coordinate origin at the central point O_f . In this work, it is of interest to evaluate C_{nm} in (2) to represent $\bar{H}_{f_n}^i(\bar{r}_f)$ [on the left hand side of (2)] in the far zone of the feed; hence, it suffices to choose the polarization vector $\bar{e}_{nm}$ by

$$\bar{e}_{nm} = [\hat{\theta}_f F_\theta(\hat{r}_f) + \hat{\phi}_f F_\phi(\hat{r}_f)]_{\hat{r}_f = \hat{z}_{fnm}}. \quad (4)$$

The above value of $\bar{e}_{nm}$ is valid within the paraxial region of the GB in (3); this is certainly true in the far zone of the feed. Furthermore, in this far-zone representation, the axis of every m th GB beam emanating from all the N lattice points will always be parallel to each other, so that the polarization in (4) is identical for every m th GB launched from each of the N lattice points.

The coefficients of expansions C_{nm} in (2) can be found in a relatively simple manner, since the feed pattern is known, using the point matching technique so that (2) can be reduced to a set of linear simultaneous equations that can be solved using standard matrix inversion. For reflector feed antenna applications, the $N \times M$ total coefficients C_{nm} are relatively small in number so that the size of the matrix to be inverted is not large, thereby making it very efficient to find C_{nm} . Alternatively, the C_{nm} can be found via a least squares approach.

For the present application, it is important to note that one is concerned here with the development of a GB expansion for representing the radiation pattern of a feed antenna in which the feed is always relatively small in size as compared to the reflector which it illuminates so that it is essentially in the far zone of the feed; hence, for this application, and even for a feed array which can be used to illuminate a reflector under similar conditions, it suffices to have just a single lattice point (i.e., $N = 1$ only) at O_f in Fig. 2 from where M rotationally symmetric GBs are launched. Thus, there are a total of only $M \times N = M \times 1 = M$ GBs in this case. The case where a large feed array needs to be decomposed into subarrays is the topic of a separate paper which also addresses the problem of reflector feed array synthesis in contoured beam applications [23]. Before determining the number of GBs, one can define a cone of half angle $\theta_{\max}$ whose apex is at O_f . This cone half angle $\theta_{\max}$ is measured from the cone axis which is also the feed axis (or feed boresight); it is chosen to adequately cover the solid angle

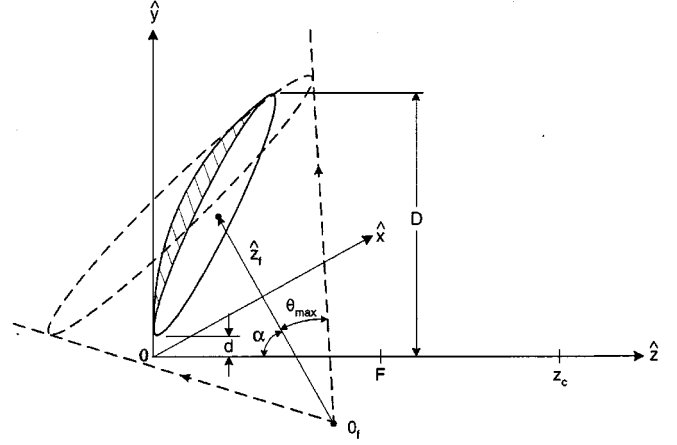


Fig. 3. Cone of half-angle $\theta_{\max}$ which encloses the reflector with a feed at O_f . The reflector has an offset as shown by the distance d . The projected reflector diameter is D , and the reflector focal length is F .

subtended by the reflector edge from the point O_f as shown in Fig. 3. It is important to note that the feed pattern needs to be represented in terms of the GB expansion only within this cone of half angle $\theta_{\max}$ to calculate the fields scattered from the reflector surface; thus it is not necessary to find a GB expansion for the feed radiation outside this cone. It is convenient next to introduce the mapping

$$\theta_x = \theta_f \cos \phi_f, \quad \theta_y = \theta_f \sin \phi_f. \quad (5)$$

It is clear from (5) that $\theta_f = \sqrt{\theta_x^2 + \theta_y^2}$ and that $\cos \phi_f = (\theta_x/\theta_f)$ with $\sin \phi_f = (\theta_y/\theta_f)$. Since ϕ_f is undefined as $\theta_f \rightarrow 0$ it is convenient to let $\phi_f = 0$ in this case. The angular inter-beam separation is chosen to be a constant between all the GBs. In particular, this condition can be satisfied if the interbeam angular spacings $\Delta\theta_x$ and $\Delta\theta_y$ in the (θ_x, θ_y) space satisfy the following:

$$\Delta\theta_y = \Delta\theta, \quad \Delta\theta_x = \Delta\theta \cdot \sin(\pi/3), \quad (6)$$

where $\Delta\theta$ is a constant. The axial direction of each of the M GBs launched from the single point O_f is shown by dots in the spherically angular (θ_x, θ_y) space of Fig. 4. The M beams in Fig. 4 can be regrouped into a set of (p, q) beams in the (θ_x, θ_y) space. Thus, $m \rightarrow (p, q)$, and for each m th GB ($m = 1, 2, \dots, M$) there is a corresponding pair of numbers (p, q) . According to Fig. 4, one allows p to vary from $-P$ to P along the θ_x axis where

$$P = \text{Int} \left[\frac{\theta_{\max}}{\Delta\theta} \right] + 1, \quad -P \leq p \leq P \quad (7)$$

$$\theta_x^{pq} = p\Delta\theta_x = \theta_x^p \quad (8)$$

and $\text{Int}[\cdot]$ denotes the integer portion of the argument. For a given θ_x^p , the number of GBs in the θ_y direction is

$$Q_p = \text{Int} \left[\frac{\sqrt{(\theta_{\max})^2 - (\theta_x^p)^2}}{\Delta\theta_y} \right] + 1. \quad (9)$$

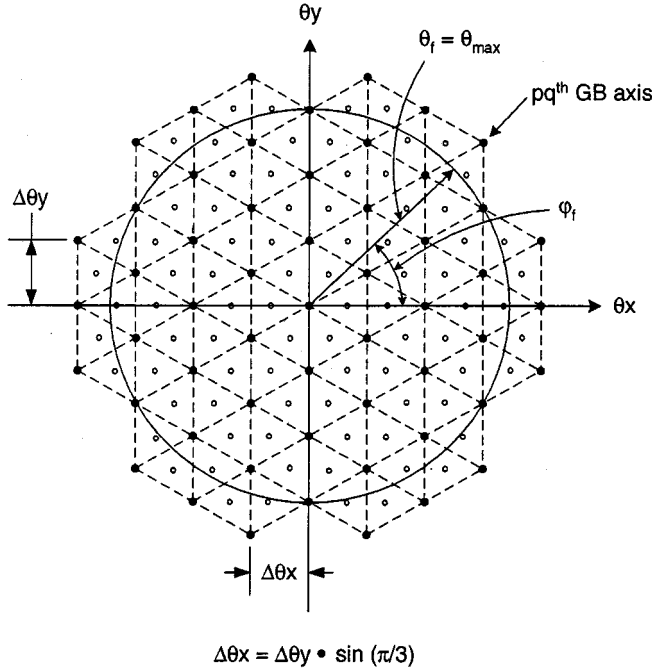


Fig. 4. Dots indicate the directions of the M GBs in the space (θ_x, θ_y) . This (θ_x, θ_y) space covers the region on the reflector surface and its neighborhood which lies within the cone of half angle $\theta_{\max}$ of Fig. 3.

The index q in the θ_y direction is chosen such that if p is an even number then

$$\theta_y^{pq} = q\Delta\theta_y, \quad -Q_p \leq q \leq Q_p \quad (10)$$

otherwise

$$\theta_y^{pq} = (q - 0.5)\Delta\theta_y, \quad -Q_p + 1 \leq q \leq Q_p. \quad (11)$$

Using (8) and (10) in $\theta_f = \sqrt{\theta_x^2 + \theta_y^2}$ yields

$$\theta_f^{pq} = \sqrt{(\theta_x^{pq})^2 + (\theta_y^{pq})^2}. \quad (12)$$

Up to this point, it is assumed that $\Delta\theta$ is known so that M is therefore also known for a given $\theta_{\max}$. With this information, one can solve for the coefficients $C_{nm} = C_{1m}$ in (2) for the $n = N = 1$ case of interest using the point matching technique as discussed earlier.

For the $N > 1$ case, one is referred to Fig. 1 which indicates how M GBs are launched from N lattice points. As indicated earlier, one may need $N > 1$ if the reflector is not almost in the far zone of a feed array. In such situations, $N = 4$ usually suffices. Points of the N rectangular lattice may correspond to the phase centers of N subapertures into which the original aperture $|x| \leq a/2$; $|y| \leq b/2$ is discretized. This situation can be treated in exactly the same fashion as done for the single lattice ($N = 1$) case, because the M GBs launched from each n th lattice point ($1 \leq n \leq N$) in Fig. 1 are required to represent only the radiation field which is generated by each of the corresponding n th subaperture containing only the n th lattice point. Furthermore, one can define a $\theta_{\max}$, n for each n th lattice point

in the same way as $\theta_{\max}$ defined in Fig. 4 for the $N = 1$ case, except that O_f in Fig. 3 must now be replaced by O_{fn} of Fig. 2. The procedure to find the amplitudes of the M GBs launched from the n th lattice is then exactly analogous to that done above for the $N = 1$ case.

It now remains to present some rules for determining $\Delta\theta$ and the parameter b in (3) which indicates the size of the GB waist. Here, b is the same for each of the identical, rotationally symmetric GBs at launch. It was determined through numerical experimentation in [6], [27] that

$$\Delta\theta \approx \sqrt{\frac{8 \ln 4}{3kb}} \quad (13)$$

with $k = 2\pi/\lambda$ (where $\lambda =$ free space wavelength). This result was obtained in two dimensions (2-D) by first assuming that the radiation pattern function (in this case, F_θ and F_ϕ) is slowly varying with respect to $\Delta\theta$. It was also assumed for the 2-D case that for a given GB in the far field, only the two nearest or adjacent GBs overlap significantly in the vicinity of the axis of the given beam. However, the condition in (13) is found, again via numerical experimentation, to be also sufficient for guaranteeing an adequate sampling of the feed pattern for the three-dimensional (3-D) case of interest in this work. That this condition in (13) should work well for 3-D is also due to the choice of almost identical interbeam angular spacing as illustrated in Fig. 4 where it is seen that the GBs along diametrically opposite sides of any hexagonal lattice resemble three overlapping GBs of the 2-D case mentioned earlier. The choice of the relationship between $\Delta\theta$ and b defined in (13) was found to give excellent reproduction of the radiation pattern when used in the GB expansion.

It therefore remains to determine b . Firstly, one notes that the half width w of each of the rotationally symmetric GBs launched from the feed plane has a spot area $A_0 = \pi w^2$ at any axial distance d_f from the launch point, where for the special GB whose axis is along $\hat{z}_f$ (see Fig. 3) the w is given by

$$w = \sqrt{\frac{2(d_f^2 + b^2)}{kb}}. \quad (14)$$

A solution for b in terms of w is obtained from (14) as

$$b = \left(\frac{kw^2}{4}\right) \pm \sqrt{\left(\frac{kw^2}{4}\right)^2 - d_f^2} \quad (15)$$

where it is generally more advantageous to pick the smaller of the two b values in (15) to ensure a smoother GB representation of the feed pattern with fewer beams as is evident from (13). The result in (15) provides a real valued b only if

$$\frac{kw^2}{4} \geq d_f. \quad (16)$$

From (14) and (16), it follows that the spot size area A_0 is given by

$$A_0 = \pi w^2 \geq 4\pi d_f/k. \quad (17)$$

It was also indicated in [1, Appendix B] that a simple closed-form approximation is available to accurately locate the complex point of diffraction associated with the diffraction of an incident GB by the edge of the reflector which it illuminates; this simple closed-form result is valid provided $\sin \phi_{1se} < 1/4$ (with ϕ_{1se} defined in [1], and also later in Section III of this paper) which leads to

$$A_0 \leq \frac{\pi D^2}{16}. \quad (18)$$

In (18), D is the diameter of the circular boundary which circumscribes the projected aperture projected in any $z = \text{constant}$ plane by the reflector as for example in Fig. 3. Clearly A_0 must therefore satisfy both the above conditions, namely the ones in (17) and (18), simultaneously. Therefore, combining (17) and (18) yields

$$4\pi d_f/k \leq A_0 (= \pi w^2) \leq \frac{\pi D^2}{16}.$$

or

$$\frac{4d_f}{k} \leq w^2 \leq \frac{D^2}{16}. \quad (19)$$

Consequently, one may pick a convenient value of w which satisfies (19); this value of w in turn provides a choice for the value of b via (15) and hence also for $\Delta\theta$ via (13) thereby specifying each GB completely at launch. For more complex feed patterns, one can always decrease $\Delta\theta$ from its value in (13) while keeping b the same as in (15); this allows the use of more GBs that may become necessary in this case to more accurately reproduce the complexity in the pattern. In general, the conditions in (13) and (15) are sufficient for most types of simple feeds.

III. SCATTERING OF EACH GB IN THE EXPANSION FOR THE FEED RADIATION WHICH ILLUMINATES THE REFLECTOR SURFACE

Once the far-zone magnetic field radiated from the feed antenna is expanded in terms of a set of GBs as shown in Section II, then each GB which is launched from the feed plane in this expansion now illuminates the reflector from where it is scattered. It is noted that the reflector may not be fully in the far zone of the feed [as in the case where $N > 1$ in (2)] even though the GB expansion is obtained from a knowledge of the far zone feed radiation pattern, and the GB expansion thus obtained remains valid even within the near zone (but excluding the extreme near zone of the feed). The net field scattered by the reflector when it is illuminated by the feed antenna is thus represented as a superposition of the scattered fields produced by each GB when they become incident on the reflector. The explicit results for the scattering of an arbitrary GB by a smooth, electrically large reflecting surface containing an edge were developed in [1]; these results which were obtained via an asymptotic evaluation of the PO integral for the scattered field are summarized below for convenience.

In [1], it was assumed that the surface of the original reflector could be approximated by a local paraboloid at the point O_m

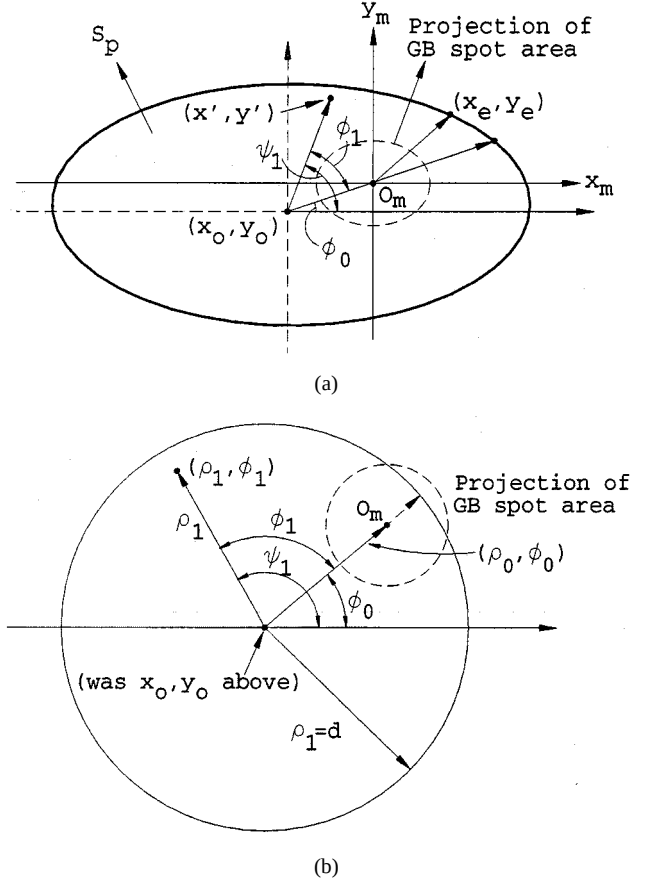


Fig. 5. (a) The projected area of the parabolic surface S on the $z_m = 0$ plane and (b) its mapping to the (ρ_1, ψ_1) coordinate system. The quantities ρ_0 , ϕ_0 and ϕ_1 are explicitly shown.

where the m th incident GB axis intersects the reflector surface (or its mathematical extension past the edge). The local paraboloid at O_m is given by

$$z_m \approx -\frac{1}{2} \left(\frac{x_m^2}{R_1} + \frac{y_m^2}{R_2} \right) \quad (20)$$

in which $\hat{z}_m$ is normal to the original surface at O_m , and $(\hat{x}_m, \hat{y}_m)$ are the original principal surface directions at O_m , respectively, as in Fig. 5. Also, $\hat{n}_r$ is a unit vector normal to the plane containing the edge with the principal radii of curvatures (R_1, R_2) , given by

$$\hat{n}_r = b_{11}\hat{x}_m + b_{12}\hat{y}_m + b_{13}\hat{z}_m \quad (21)$$

where (x_m, y_m, z_m) denotes the coordinate system for the local parabolic surface at O_m . It is assumed that the incident magnetic field of the m th rotationally symmetric GB incident at O_m is given by [1]

$$\begin{aligned} \bar{H}_m^i(\bar{r}_i) = & \bar{H}_m^i(O_m) \sqrt{\frac{\text{Det}[Q^i(z_{fm})]}{\text{Det}[Q^i(0)]}} \\ & \cdot e^{-jk(z_{fm} + 1/2[\xi_m^i]^T Q^i(z_{fm})[\xi_m^i])}, \\ [\xi_m^i] = & \begin{bmatrix} x_{fm} \\ y_{fm} \end{bmatrix} \end{aligned} \quad (22)$$

where $\bar{r}_{nm}^i = (x_{fm}, y_{fm}, z_{fm})$ is the position vector in the incident GB coordinate system which is also centered at O_m and $\bar{H}_m^i(0)$ is the incident magnetic field at O_m , i.e., at $\bar{r}_i = (0, 0, 0)$. The $Q_{11}^i, Q_{12}^i = Q_{21}^i$, and Q_{22}^i are the elements of the GB complex curvature (or phase) matrix $Q^i(0)$ at $z_{fm} = 0$. In particular, (22) reduces at $z_{fm} = 0$ to the following:

$$\bar{H}_m^i(O_m) = \bar{e}_m \frac{jb}{d_m + jb} e^{-jkd_m} \quad (23)$$

with

$$Q^i(0) = \begin{bmatrix} Q_{11}^i & Q_{12}^i \\ Q_{12}^i & Q_{22}^i \end{bmatrix} = \begin{bmatrix} \frac{1}{d_m + jb} & 0 \\ 0 & \frac{1}{d_m + jb} \end{bmatrix} \quad (24a)$$

and

$$[Q^i(z_{fm})]^{-1} = [Q^i(0)]^{-1} + z_{fm} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (24b)$$

It is clear that d_m appearing in (23) and (24a) represents the distance $|\bar{O}_f \bar{O}_m|$ associated with the m th GB. The transformation matrix between the coordinates fixed in the incident GB and the local surface coordinate system both of which are centered at O_m is

$$\begin{bmatrix} x_{fm} \\ y_{fm} \\ z_{fm} \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \begin{bmatrix} x_m \\ y_m \\ z_m \end{bmatrix}. \quad (25)$$

Clearly, the 3×3 transformation matrix in (22) is known. The scattered field $\bar{P}^s$ which is produced when a GB is incident on the local parabolic surface can be written asymptotically for large k as

$$\bar{P}^s(\bar{r}) \cong \bar{P}_r(\bar{r}) \cdot T(S_a) + \bar{P}_d(\bar{r}) \quad (26)$$

where $\bar{P}_r(\bar{r})$ represents the reflected field contribution to the scattered field which actually would be the only contribution if the edge were absent. Also, $\bar{P}_d(\bar{r})$ represents the edge diffracted field contribution, and $T(S_a)$ is a transition function that is obtained via uniform asymptotics, which provides a modification to the reflected component of the field resulting from the presence of the edge. Indeed, when the edge is present, only a part of the incident GB is reflected from the surface, and the rest is diffracted by the edge; thus, the effect of the surface reflection is modified by the edge and the extent of this modification depends on the proximity of O_m to the edge as measured by the parameter S_a of the transition function $T(S_a)$. It is noted that $\bar{P}^s(\bar{r})$ denotes either the scattered electric field or the scattered magnetic field according to the function $\bar{F}(x_m, y_m)$ which is defined below. The individual field components of $\bar{P}^s$ are summarized in detail next.

A. Reflection Contribution $\bar{P}_r(\bar{r})$

The reflected field $\bar{P}_r(\bar{r})$ is given by [1]

$$\bar{P}_r(\bar{r}) = -\pi \bar{F}(x_s, y_s) \frac{1}{\sqrt{-jk a_1}} \frac{1}{\sqrt{-\frac{jk(a_0 a_1 - c^2)}{a_1}}} \cdot e^{jk f(x_s, y_s)} \quad (27)$$

where

$$\bar{F}(x_s, y_s) = \begin{cases} \frac{jkZ_0}{2\pi R} \left[\hat{R} \times \hat{R} \times \left(\hat{z}_m + \hat{x}_m \frac{x_s}{R_1} + \hat{y}_m \frac{y_s}{R_2} \right) \right. \\ \quad \left. \times \bar{H}_{am}^i(\bar{r}'_m) \right], & \text{for } \bar{E}(\bar{r}_m) \\ \frac{-jk}{2\pi R} \left[\hat{R} \times \left(\hat{z}_m + \hat{x}_m \frac{x_s}{R_1} + \hat{y}_m \frac{y_s}{R_2} \right) \right. \\ \quad \left. \times \bar{H}_{am}^i(\bar{r}'_m) \right], & \text{for } \bar{H}(\bar{r}_m). \end{cases} \quad (28)$$

and

$$\bar{H}_{am}^i(\bar{r}'_m) = \bar{H}_m^i(O_m) \sqrt{\frac{\text{Det}[Q^i(z_{fm})]}{\text{Det}[Q^i(0)]}}. \quad (29)$$

In (23), R denotes the distance between the observation point and the complex stationary or reflection point on the complex extension of the surface at $\bar{r}'_m = (x_s, y_s, z_s)$ in the coordinate system at O_m , and $\hat{R}$ denotes its complex direction pointing toward the observer at $\bar{r} = (x_m, y_m, z_m)$ also in the O_m system. It is noted that the quantities (x_s, y_s, z_s) and $f(x_s, y_s)$ are given in closed form as

$$x_s = \frac{a_1 b_0 - b_1 c}{2(a_0 a_1 - c^2)} \quad (30a)$$

$$y_s = \frac{a_0 b_1 - b_0 c}{2(a_0 a_1 - c^2)} \quad (30b)$$

$$z_s = -\frac{1}{2} \left(\frac{x_s^2}{R_1} + \frac{y_s^2}{R_2} \right) \quad (30c)$$

and

$$f(x_s, y_s) = -r + \frac{a_1 b_0^2 + a_0 b_1^2 - 2b_1 b_0 c}{4(a_0 a_1 - c^2)} \quad (31)$$

in which the coefficients a_0, b_0, a_1, b_1 , and c are

$$a_0 = \frac{1}{2} \left\{ \frac{1}{R_1} \left(\frac{z_m}{r} - c_{33} \right) + \frac{1}{r} \left(1 - \frac{x_m^2}{r^2} \right) + c_{11}^2 Q_{11}^i + c_{21}^2 Q_{22}^i + 2c_{11} c_{21} Q_{12}^i \right\} \quad (32)$$

$$b_0 = \left(\frac{x_m}{r} - c_{31} \right) \quad (33)$$

$$a_1 = \frac{1}{2} \left\{ \frac{1}{R_2} \left(\frac{z_m}{r} - c_{33} \right) + \frac{1}{r} \left(1 - \frac{y_m^2}{r^2} \right) + c_{12}^2 Q_{11}^i + c_{22}^2 Q_{22}^i + 2c_{12}c_{22}Q_{12}^i \right\} \quad (34)$$

$$b_1 = \left(\frac{y_m}{r} - c_{32} \right) \quad (35)$$

$$c = \frac{1}{2} \left\{ c_{11}c_{12}Q_{11}^i + c_{21}c_{22}Q_{22}^i + (c_{11}c_{22} + c_{21}c_{12})Q_{12}^i - \frac{x_my_m}{r^3} \right\} \quad (36)$$

$$r = \sqrt{x_m^2 + y_m^2 + z_m^2}. \quad (37)$$

B. Edge Diffraction Contribution $\bar{P}_d(\bar{r})$

The edge diffracted field $\bar{P}_d(\bar{r})$ is given by [1]

$$\bar{P}_d(\bar{r}) = \bar{F}(x_{se}, y_{se}) ABd \frac{\sqrt{2\pi}}{\sqrt{jk g_\phi''(\phi_{1se})}} \frac{1}{kg_\rho'(d)} e^{jkf(x_{se}, y_{se})} \quad (38)$$

where

$$A = \left| \sqrt{-\frac{2R_1}{b_{13}}} \right| \quad (39)$$

$$B = \left| \sqrt{-\frac{2R_2}{b_{13}}} \right| \quad (40)$$

$$d = \left| \sqrt{\left| x_e - \frac{b_{11}^2 R_1}{2b_{13}} - \frac{b_{12}^2 R_2}{2b_{13}} \right|} \right| \quad (41)$$

and

$$f(x_{se}, y_{se}) = f(x_s, y_s) - a_0(x_{se} - x_s)^2 - a_1(y_{se} - y_s)^2 - 2c(x_{se} - x_s)(y_{se} - y_s). \quad (42)$$

The coordinates (x_{se}, y_{se}) above pertain to a complex saddle point associated with edge diffraction. It is convenient to employ polar coordinates in which to express (x_{se}, y_{se}) ; in particular, let

$$x_e = x_o + Ad \cos \psi_1 \quad (43a)$$

$$y_e = y_o + Bd \sin \psi_1. \quad (43b)$$

At this saddle point, the values of (x_e, y_e) and ψ_1 are denoted by (x_{se}, y_{se}) and ψ_{1se} , respectively, or

$$x_{se} = x_o + Ad \cos \psi_{1se} \quad (44a)$$

$$y_{se} = y_o + Bd \sin \psi_{1se} \quad (44b)$$

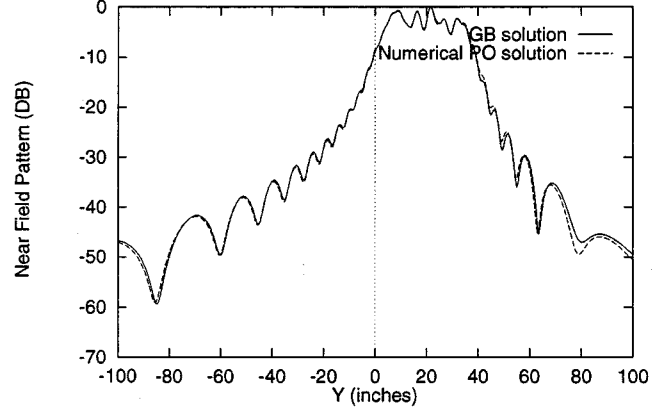


Fig. 6. Near-field pattern at 5 GHz of an offset parabolic reflector illuminated by a Huygens source located at the focus where ($D = 40''$, $F = 36''$, $d = 0''$, $z_c = 40''$). Forty-eight GBs are used.

where $\psi_{1se} = \phi_{1se} + \phi_0$ as in Fig. 6(a) and (b) with

$$\rho_0 = \sqrt{\frac{x_o^2}{A^2} + \frac{y_o^2}{B^2}} \quad (45)$$

$$\cos \phi_0 = -\frac{x_o}{A\rho_0} \quad (46)$$

$$\sin \phi_0 = -\frac{y_o}{B\rho_0} \quad (47)$$

and

$$x_o = \frac{b_{11}R_1}{b_{13}} \quad (48)$$

$$y_o = \frac{b_{12}R_2}{b_{13}}. \quad (49)$$

The derivatives $g_\rho'(d)$ and $g_\phi''(\phi_{1se})$ are explicitly given in [1] by

$$g_\rho'(d) = [-2a_0(x_{se} - x_s) - 2c(y_{se} - y_s)](A \cos \psi_{1se}) + [-2a_1(y_{se} - y_s) - 2c(x_{se} - x_s)](B \sin \psi_{1se}) \quad (50)$$

and

$$g_\phi''(\phi_{1se})|_{\rho'=d} = -dg_\rho'(d) - 2d^2[a_0A^2 \sin^2 \psi_{1se} - 2cAB \sin \psi_{1se} \cdot \cos \psi_{1se} + a_1B^2 \cos^2 \psi_{1se}]. \quad (51)$$

It is noted that the saddle point ϕ_{1se} which corresponds to (x_{se}, y_{se}) in an appropriate polar coordinate system [1] is the solution of

$$H_1 \sin^2 \phi_{1se} + H_2 \sin \phi_{1se} \cos \phi_{1se} + H_3 \sin \phi_{1se} + H_4 \cos \phi_{1se} + H_5 = 0 \quad (52)$$

where H_i , $i = 1 \dots, 5$ are given by [1]

$$H_1 = 2d[(-a_0A^2 + a_1B^2) \sin 2\phi_0 + 2cAB \cos 2\phi_0] \quad (53)$$

$$H_2 = 2d[(a_0A^2 - a_1B^2) \cos 2\phi_0 + 2cAB \sin 2\phi_0] \quad (54)$$

$$H_3 \equiv 2\{a_0 A_1(x_o - x_s) + a_1 B_1(y_o - y_s) + c[A_1(y_o - y_s) + B_1(x_o - x_s)]\} \quad (55)$$

$$H_4 \equiv -2\{a_0 A_2(x_o - x_s) + a_1 B_2(y_o - y_s) + c[A_2(y_o - y_s) + B_2(x_o - x_s)]\} \quad (56)$$

$$H_5 \equiv -\frac{1}{2}H_1 \quad (57)$$

with A_1 , A_2 , B_1 , and B_2 as

$$A_1 = A \cos \phi_0 \quad (58)$$

$$A_2 = -A \sin \phi_0 \quad (59)$$

$$B_1 = B \sin \phi_0 \quad (60)$$

$$B_2 = B \cos \phi_0. \quad (61)$$

The approximate but accurate closed-form solution of (52) is shown in [1], and it will not be repeated here.

C. Uniform Transition Function $T(S_a)$

The transition function $T(S_a)$ is given by [1]

$$T(S_a) \equiv 1 - \frac{1}{2}[1 - \text{erf}(S_a)] + \frac{e^{-S_a^2}}{2\sqrt{\pi}S_a} \quad (62)$$

where erf denotes the Error function ($\text{erf}(\Omega) = (2/\sqrt{\pi}) \int_0^\Omega e^{-t^2}/dt$) and the parameter S_a depends on the complex distance between the complex saddle point for reflection and the complex saddle point for edge diffraction [see (63) at the bottom of the page], which is given in [1]. The choice for the branch of the square root is based on a Taylor expansion around the complex saddle point for reflection, i.e.,

$$\text{sign}(\text{Re}[S_a]) = \text{sign} \left\{ \text{Re} \left[\sqrt{-\frac{jk}{2}} g''_\rho(\rho_{1s})(d - \rho_{1s}) \right] \right\} \quad (64)$$

where

$$g''_\rho(\rho_{1s})|_{(x_s, y_s)} = \frac{1}{g''_\phi(\phi_{1s})} \cdot 4\rho_{1s}^2 A^2 B^2 (a_0 a_1 - c^2) \quad (65)$$

and (ρ_{1s}, ϕ_{1s}) denote a set of appropriate polar coordinate values [1] corresponding to (x_s, y_s) . Thus,

$$g''_\phi(\phi_{1s})|_{(x', y')=(x_s, y_s)} = -2\rho_{1s}[a_0 A^2 \sin^2 \psi_{1s} - 2cAB \sin \psi_{1s} \cdot \cos \psi_{1s} + a_1 B^2 \cos^2 \psi_{1s}] \quad (66)$$

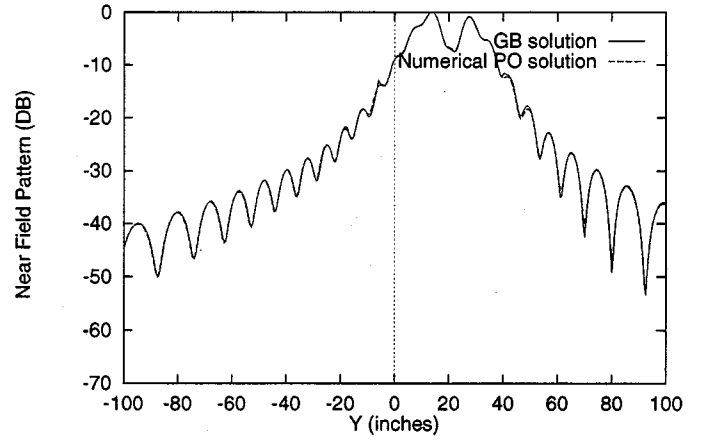


Fig. 7. Near-field pattern at 5 GHz of an offset parabolic reflector illuminated by a Huygens source located at the focus where ($D = 40''$, $F = 36''$, $d = 0''$, $z_c = 100''$). Forty-eight GBs are used.

with $\psi_{1s} = \phi_{1s} + \phi_0$. In (65) and (66), (ρ_{1s}, ψ_{1s}) is given by [1]

$$\begin{cases} \rho_{1s} = \sqrt{\frac{(x_s - x_o)^2}{A^2} + \frac{(y_s - y_o)^2}{B^2}} \\ \cos(\psi_{1s}) = \frac{(x_s - x_o)}{A\rho_{1s}} \\ \sin(\psi_{1s}) = \frac{(y_s - y_o)}{B\rho_{1s}} \end{cases} \quad (67)$$

However, it is suggested that for small S_a the Taylor expansion for S_a of (63), which is essentially shown in (64), may be employed directly to avoid any numerical error in the transition region.

IV. NUMERICAL RESULTS AND DISCUSSION

In this section, some numerical results are presented for the GB-based analysis of the fields radiated from an offset parabolic reflector illuminated by a single feed or feed array, as well as from an offset ellipsoidal reflector, and from a shaped reflector designed with a single feed to produce a contoured beam for the continental United States (CONUS). The accuracy and efficiency of the present GB approach is also discussed by comparing it with a reference solution obtained via a conventional direct numerical evaluation of the PO integral over the surface of the reflector.

The first case of interest here is the investigation of the near field pattern of an offset parabolic reflector. The geometry for this case is shown in Fig. 3 where the diameter D of the projected aperture is given by $D = 40''$ with $d = 0$, and F the focal length $= 36''$. Also $z = z_c$ is the location of the output plane, and the near-field patterns are in the $x = 0$ plane with $z = z_c$. The feed antenna is a simple Huygens source located at the focus, and the frequency is 5 GHz. The Huygens source is tilted so that its boresight (or axis) makes an angle α in the $x = 0$ plane given by $\alpha = 30^\circ$, and its electric current moment lies in the $x = 0$ plane. Figs. 6 and 7 show the near field patterns of

$$\begin{aligned} S_a &= \sqrt{jk[f(x_s, y_s) - f(x_{se}, y_{se})]} \\ &= \sqrt{jk[a_0(x_{se} - x_s)^2 + a_1(y_{se} - y_s)^2 + 2c(x_{se} - x_s)(y_{se} - y_s)]}. \end{aligned} \quad (63)$$

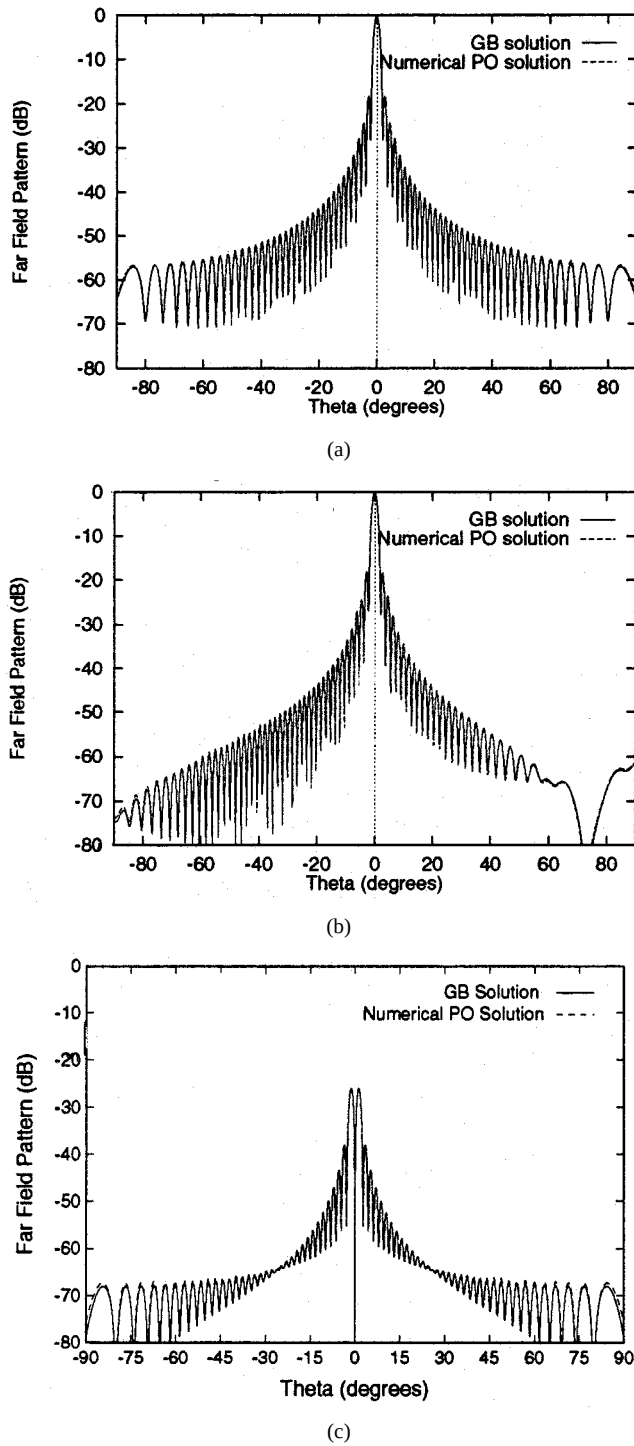


Fig. 8. Co-polar and cross-polar far-field pattern at 10 GHz of an offset parabolic reflector used in Fig. 6 when it is illuminated by a Huygens source located at the focus. 32 GBs are used. (a) $\phi = 0^\circ$ plane (co-polar). (b) $\phi = 90^\circ$ plane (co-polar). (c) $\phi = 0^\circ$ plane (cross-polar). Note that there is no cross-polar field in the $\phi = 90^\circ$ plane.

this reflector at two different output planes. In particular, Fig. 6 shows the pattern in the output plane located at $z_c = 40''$ which is relatively very close to the reflector, whereas Fig. 7 shows the pattern in the output plane located at $z_c = 100''$ which is still in the near zone but much farther from the reflector. Both Figs. 6 and 7 show very good agreement with the reference solution. It is noted that the present GB approach is based on the asymptotic

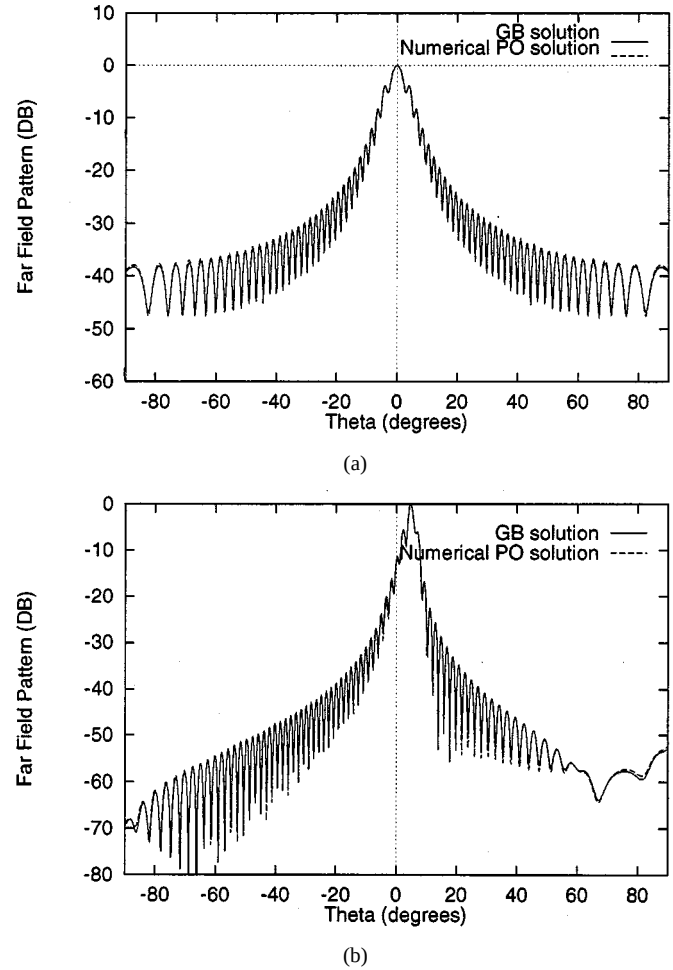


Fig. 9. Far-field pattern at 10 GHz of an offset parabolic reflector used in Fig. 6 when it is illuminated by a defocused Huygens source located at $(0., 0., 30'')$. Thirty-two GBs are used. (a) $\phi = 0^\circ$ plane. (b) $\phi = 90^\circ$ plane.

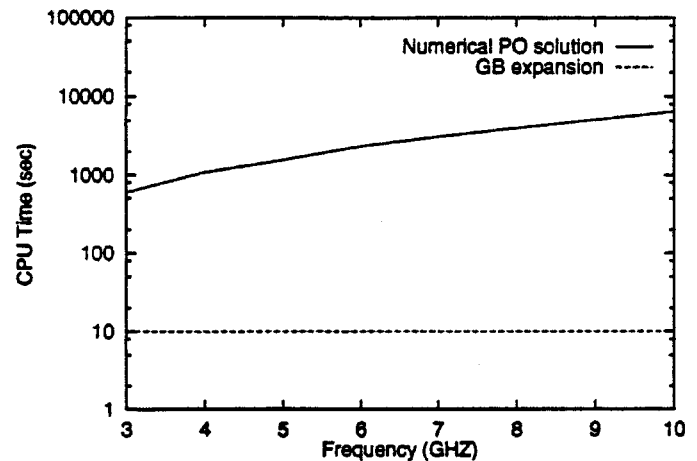


Fig. 10. CPU time required on a silicon graphics workstation for the reference numerical PO integration-based far field computation compared with that based on GB expansion. The time shown is for computing 1802 pattern points for the reflector used in Figs. 6–9. Note that the vertical scale is logarithmic.

development in [1] that employs a Fresnel approximation within the PO radiation integral for the scattered field produced by each incident GB. It is expected that some errors will occur if the field point is not in the Fresnel zone over the spot area on the

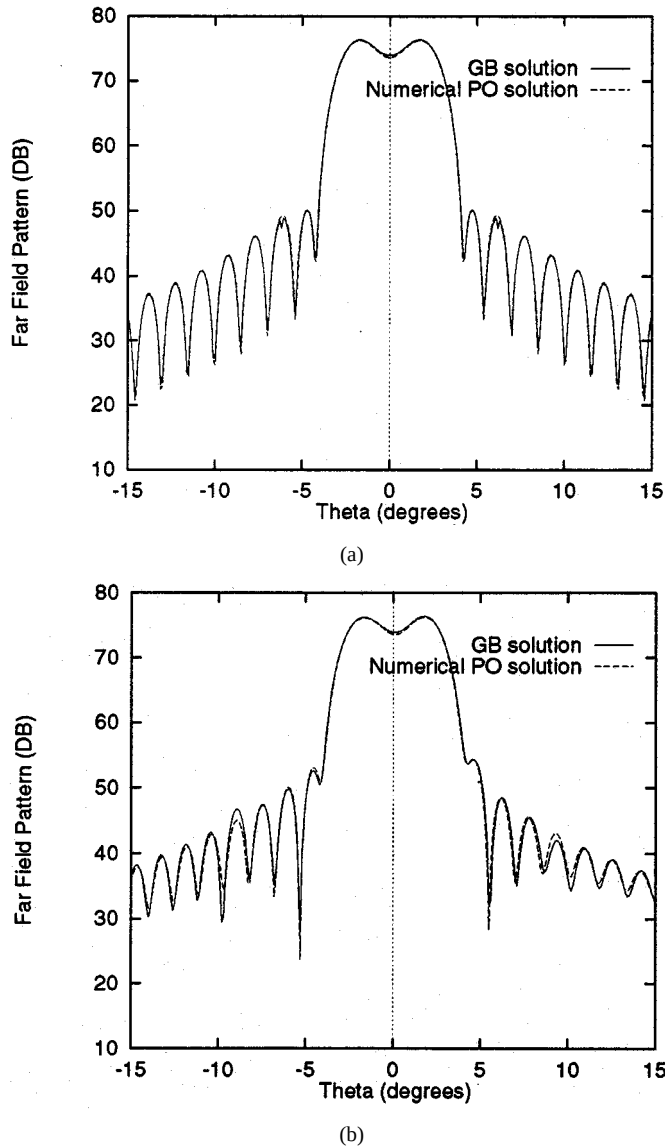


Fig. 11. Far-field pattern at 121.803 GHz for the parabolic reflector utilized in Figs. 6–9 except that it is illuminated by a 9×9 element rectangular array with an interelement spacing of a half wavelength. The center of the array is at the focus of the parabolic reflector. (a) $\phi = 0^\circ$ plane. (b) $\phi = 90^\circ$ plane. Eighty-nine GBs are used.

reflector which is made by each GB in the feed expansion. Fig. 7 shows the improvement in accuracy when all the field points are properly located in the Fresnel zone at a larger distance $z_c = 100''$ as compared to $z_c = 40''$. Also, in the present application, the spot area of the incident GB on the surface of the reflector is assumed to be small compared with the total surface area of the offset reflector as discussed in Section II. The total number of GBs used is 48 in Figs. 6 and 7.

The second case of interest is the computation of the far-field pattern for the same reflector as the one used in Figs. 6 and 7. Again, the Huygens source feed is at the focus and its axis is tilted at an angle α in the $x = 0$ plane as before for this reflector whose far zone pattern is shown in Fig. 8 at 10 GHz. When the feed is defocused, the far-zone pattern changes to that in Fig. 9. It is noted that, being an asymptotic high-frequency solution, the accuracy of the GB based solution increases with frequency.

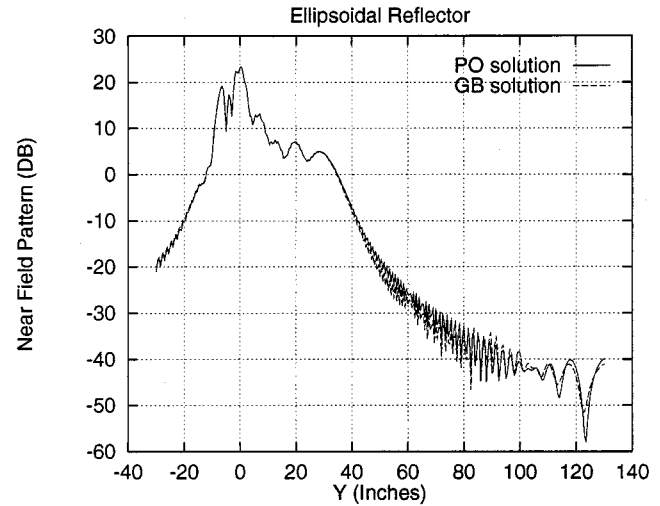


Fig. 12. Near-field pattern at 11.803 GHz of an offset ellipsoidal reflector illuminated by a Huygen's source placed on the axis at a distance of 70 in from the origin (see Fig. 3). Note that ($D = 80$ in, $d = 0$ in, $z_c = 100$ in). Approximately 50 GBs are used.

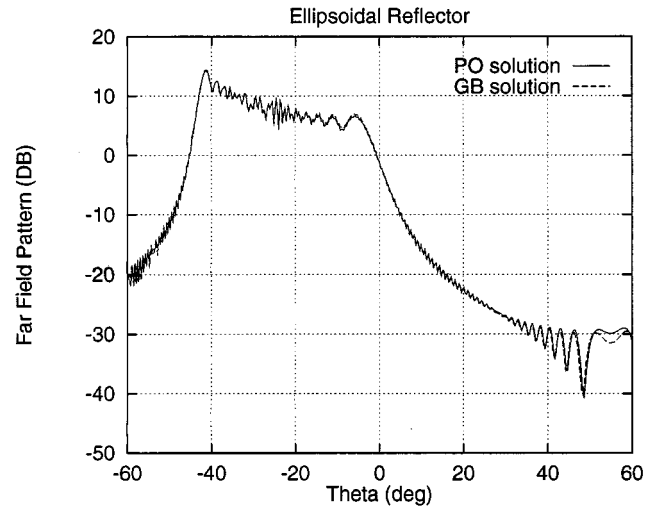


Fig. 13. Far-field pattern at 11.803 GHz of an offset ellipsoidal reflector illuminated by a Huygen's source placed on the axis at a distance of 70 in from the origin (see Fig. 3). Note that ($D = 80$ in, $d = 0$ in) for this case, which is the same as for Fig. 12 except that this figure depicts the far-field pattern. Approximately 50 GBs are used.

For the near-field case in Figs. 6 and 7, a total of 48 GBs is used as indicated above; in the far zone one can further reduce the number of GBs to 32 as in Figs. 8 and 9 and this number of GBs stays the same even when the frequency is doubled. It is noted that Fig. 8(c) indicates the cross-polar field radiated in the far zone by the reflector. The CPU time for the GB approach is approximately 10 s on a Silicon Graphics workstation for a total of 1802 field point samples used in each of the plots of Figs. 8 and 9; on the other hand, the CPU time for the reference numerical PO solution increases rapidly with frequency as shown in Fig. 10, and is typically given in terms of hours. The numerical PO integration code used here, to provide the reference solution, employs a surface discretization in which the surface elements are 0.1×0.1 square wavelengths in size. It is possible to improve the efficiency of this numerical PO code somewhat;

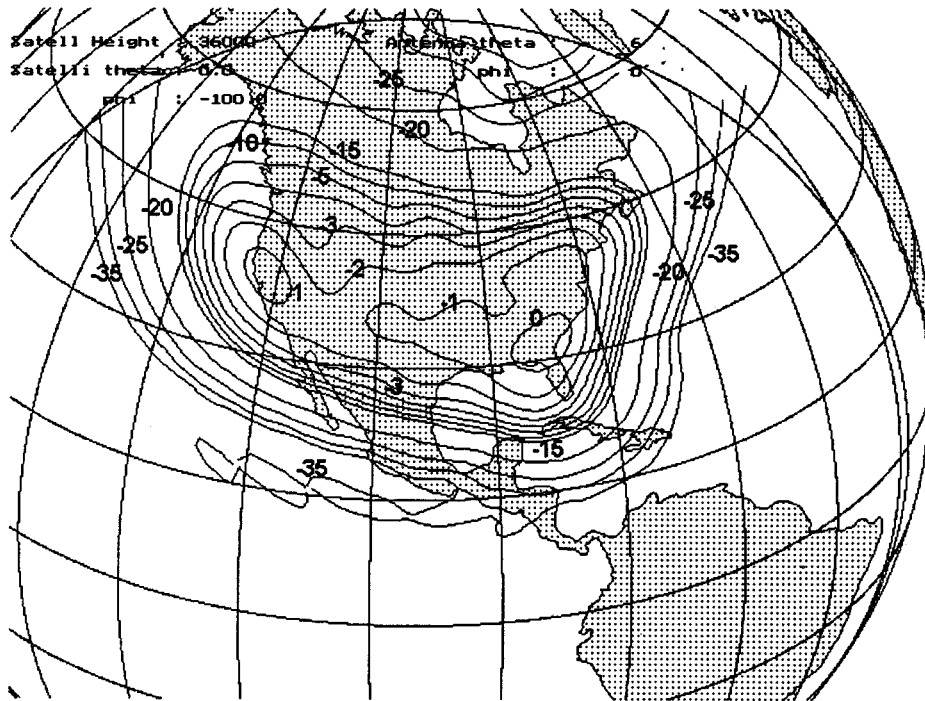


Fig. 14. Normalized co-polarization contours based on the GB approach for CONUS coverage by a shaped concave reflector with a $(\cos \theta_f)^l$ feed pattern at 12 GHz with $l = 18.51$. Approximately 200 GBs were used.

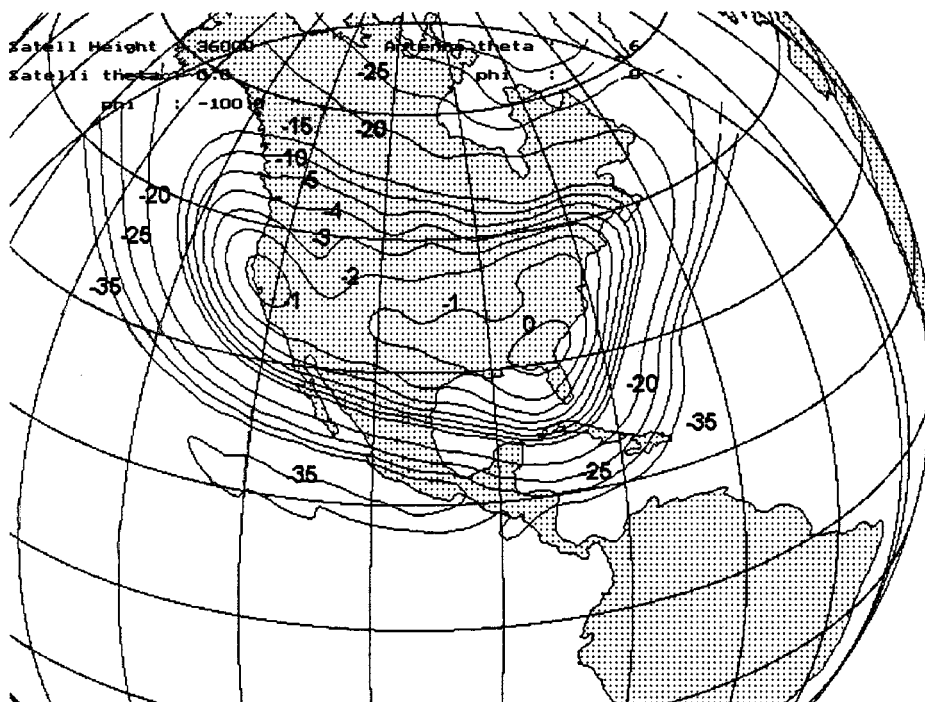


Fig. 15. Normalized co-polarization gain contours based on the numerical PO integration approach for the same shaped reflector case as in Fig. 14.

however, even with that improvement, the numerical PO code will still require significantly much larger computational time, especially for higher frequencies, than the novel GB method of solution presented here.

The third case considered here is the computation of the far field pattern of the same parabolic reflector as the one utilized in

the computations of Figs. 6–9, except that it is now fed by a 9×9 element planar square antenna array whose center is at the focus and whose boresight points toward the center of the reflector at an angle α as in the case of Figs. 7–9. Each radiating array element consists of Huygens source identical to that employed for the calculations in Figs. 6–9. The interelement array spacing

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